

Lab 3, Act 3

From counts to a continuous measurement

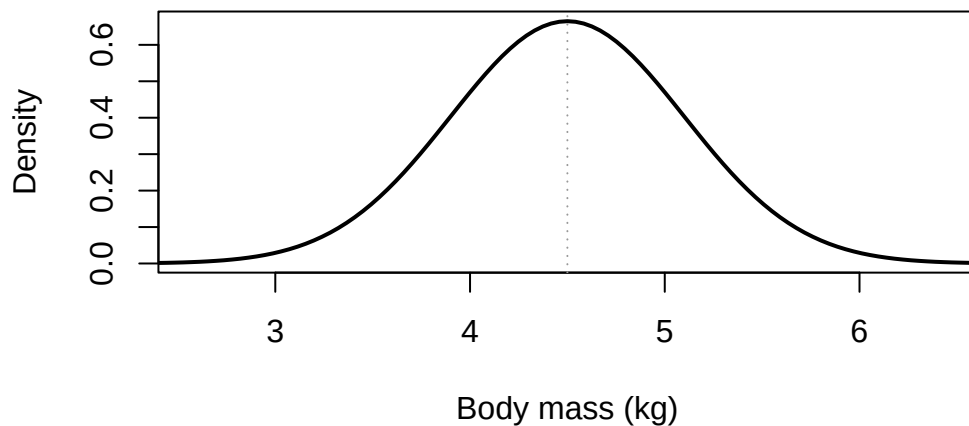
NRES 746 – name: _____ group: _____

Picking up where we left off

So far this lab has treated detections as discrete counts (0, 1, 2, ... detections). But some of the quantities the field team records are continuous. Suppose adult body mass for this carnivore, estimated from reference objects visible in camera-trap images, is well described by a normal distribution with mean $\mu = 4.5$ kg and standard deviation $\sigma = 0.6$ kg.

Part 1: Reading a density curve (individual, ~10 min)

Below is the probability density function (pdf) for this normal distribution.



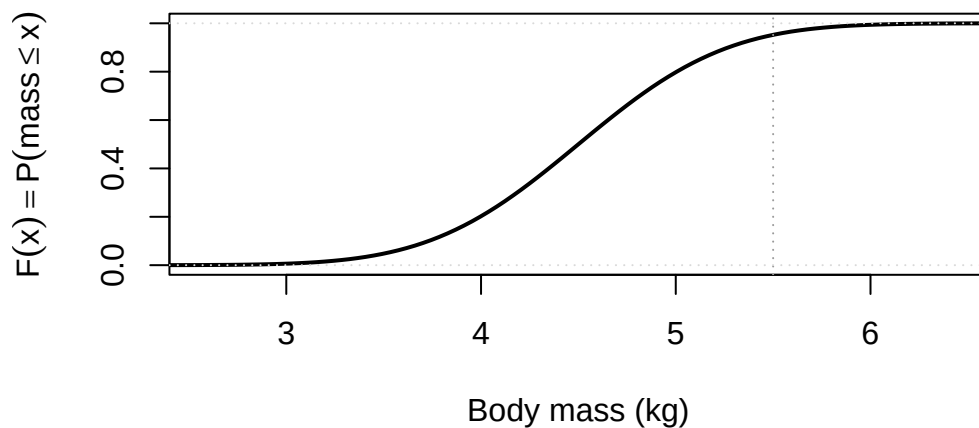
1a. Shade it by hand. On the curve above, shade the region that represents $P(4.5 < \text{mass} < 5.5)$.

1b. What is $P(\text{mass} = 4.5)$ exactly? Why – what's different about a single point on a continuous curve versus an interval?

1c. The height of the curve at mass = 4.5 is not zero (in fact, it's the highest point on the whole curve). If that height isn't a probability, what does it actually represent?

Part 2: Reading a CDF (individual, ~5-8 min)

Below is the cumulative distribution function (CDF), $F(x) = P(\text{mass} \leq x)$, for the same distribution.



2a. Using the CDF plot, estimate $P(\text{mass} > 5.5)$ (read off $F(5.5)$ as best you can, then subtract). Record your estimate – you don’t need to be precise, just in the right neighborhood.

Part 3: Standardize and look it up (individual, ~10-12 min)

Graph-reading only gets you so far. The **standard normal table** (Table 1, at the end of this Act) gives exact values of $\Phi(z) = P(Z \leq z)$ for the *standard* normal distribution ($\mu = 0$, $\sigma = 1$). To use it for a normal distribution with a different mean and standard deviation, first convert your value of interest to a **z-score**:

$$z = \frac{x - \mu}{\sigma}$$

This rescales x into “number of standard deviations away from the mean,” which is exactly what the table expects.

3a. Compute the z-score for $x = 5.5$ kg.

3b. Look up $\Phi(z)$ for the z-score you just computed (round z to the nearest 0.01 if needed), then use it to compute $P(\text{mass} > 5.5)$. How close was this to your graph-reading estimate from 2a?

3c. Now go the other direction. Search the *body* of the table for the value closest to 0.90, and read off the z-score for that row and column – this is (approximately) the z-score marking the heaviest 10% of the population. Convert that z-score back into a body mass using $x = \mu + z\sigma$.

Group discussion and revision (~5 min)

Compare your Part 3 answers with your group. One thing worth discussing: the z-table trick in 3a-3c only works because this distribution is *normal* – what about the standardizing step, specifically, depends on that assumption?

Feel free to revise any of your answers above based on the discussion (it's fine to cross out and rewrite – I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion (“muddiest points”) or topics you want to review or dust off going forward.

Table 1: Standard normal cumulative probabilities, $\Phi(z) = P(Z \leq z)$

Row = z to one decimal place; column = second decimal place. Example: for $z = 1.67$, read the row “1.6” and the column “.07”.

	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998